

Inversion of the satellite observations of the tidally induced magnetic field in terms of 3-D upper-mantle electrical conductivity: method and synthetic tests

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SUMMARY

The interaction of the oceanic tidal flow with the Earth's main magnetic field provides a powerful natural source of electromagnetic (EM) energy suitable for suboceanic upper-mantle electrical conductivity sounding. In this paper, we have developed and tested a new frequency-domain, spherical harmonic-finite element approach to the inverse problem of global EM induction. It is set up for an effective inversion of satellite-observed tidally induced magnetic field in terms of 3-D structure of the electrical conductivity in the suboceanic upper mantle. Before proceeding to the inversion of Swarm-derived models of tidal magnetic signatures, we have performed a series of parametric studies, using the 3-D conductivity model WINTERC-e as a testbed. The WINTERC-e model has been derived using state-of-the-art laboratory conductivity measurements of mantle minerals, and thermal and compositional model of the lithosphere and upper mantle WINTERC-G. The latter model is based on the inversion of global surface waveforms, satellite gravity and gradiometry measurements, surface elevation and heat flow data in a thermodynamically self-consistent framework. Therefore, the WINTERC-e model, independent of any EM data, represents an ideal target for synthetic tests of the 3-D EM inversion. We tested the impact of the truncation degree of the spherical-harmonic expansion of the M_2 tidal signal, the effect of random noise in synthetic data and inclusion of the N_2 and O_1 tidal constituents on the ability to recover the suboceanic upper-mantle conductivity structure. We demonstrate that with suitable regularization we can successfully reconstruct the 3-D upper-mantle conductivity beneath world oceans. In the ideal noise-free case, the correlation coefficient between the target and recovered conductivity is greater than 0.8 in the 150–270 km depth range.

Key words: Tides and planetary waves; Geomagnetic induction; Inverse theory.

1 INTRODUCTION

The electromagnetic induction (EMI) is one of the fundamental geophysical methods to study the Earth's interior. Traditionally, ground-based observatory data were used to construct a 1-D conductivity profile on the global scale. However, EMI methods have evolved in the last two decades towards 3-D inversions, especially due to satellite data with nearly global coverage, and progress in numerical modelling (Kuvshinov 2015; Velínský & Knopp 2021; Kuvshinov *et al.* 2021). Recently, the traditional EM sources in the ionosphere and magnetosphere have been supplemented with the EM sources in the Earth's oceans to constrain the 1-D electrical conductivity profile in the suboceanic upper mantle, and across the lithosphere–asthenosphere boundary (Grayver *et al.* 2016, 2017).

The tidal circulation is responsible for the regular rise and fall of sea surface during the day, which is notable especially in coastal regions. It is more suitable for EMI studies than the wind- and buoyancy-driven circulation due to its less complex spatio-temporal structure. First, tidal circulation can be decomposed (at least in deep open-ocean regions where the linearized momentum equation is valid) to the contributions of specific tidal frequencies (Einšpigel & Martinec 2016). Secondly, the tidal circulation has a large-scale pattern in the 100–1000 km range (Dickey 1990; Dickey & Bidigare 2005). Thirdly, the numerical assimilative models of tidal circulation are accurate in the deep open-ocean regions (Stammer *et al.* 2014) since the tidal forcing is well understood and the circulation is predominantly depth-independent.

The tidally induced magnetic field (TIMF) is generated via the process of motional induction. The seawater is a conductive fluid which circulates in the presence of the Earth's magnetic field. The Lorentz force acting on all moving charged particles in the magnetic field generates electric currents. These currents, in turn, induce a secondary magnetic field (Sanford 1971). The magnitude of TIMF reaches units of nT at CHAMP and Swarm satellite altitudes (Velínský *et al.* 2018). Tyler *et al.* (2003) were the first who demonstrated that the TIMF can be extracted from the CHAMP satellite data. Sabaka *et al.* (2015) incorporated the tidal parametrization into the comprehensive inverse model CM5 and successfully recovered the magnetic field of the principal lunar semi-diurnal tide M_2 . The most up-to-date data products derived by Grayver & Olsen (2019) and Sabaka *et al.* (2020) contain not only the M_2 tide but also the N_2 and O_1 tides. However, there is a higher discrepancy between N_2 and O_1 TIMFs obtained by satellite data analysis, and those predicted by EMI modelling (Grayver & Olsen 2019).

The recent progress makes the TIMF a promising tool for the sounding of the Earth's interior. Grayver *et al.* (2016) recovered the 1-D conductivity model of the lithosphere and upper mantle down to 250 km depth using the M_2 constituent. Grayver *et al.* (2017) presented 1-D models of mantle conductivity obtained from the combined inversion of C_1 mantle responses to the magnetospheric ring current excitation and tidal M_2 TIMF. They demonstrated that the inclusion of TIMF improved the recovered model in the upper mantle.

Martinec *et al.* (2021) have introduced a new 3-D upper-mantle electrical conductivity model WINTERC-e. It is based on laboratory measurements of electrical conductivity of mantle minerals and melt and a thermochemical upper mantle model, WINTERC-G, derived in turn from seismic waveforms, heat flow, surface elevation, and satellite gravity data (Fullea *et al.* 2021). Moreover, Martinec *et al.* (2021) have compared the forward predictions of TIMF by the M_2 constituent at satellite altitude against actual satellite-derived tidal magnetic signatures (Grayver & Olsen 2019; Sabaka *et al.* 2020). They have concluded that a 3-D inversion of satellite-observed TIMF is feasible, and can lead to further improvements of the upper-mantle thermochemical model, especially with respect to partial melt and water content, where electrical conductivity demonstrates the largest sensitivity.

In this paper, we examine a new approach to the inverse problem to recover a 3-D upper-mantle conductivity structure beneath world oceans using TIMF. The WINTERC-e electrical conductivity model plays a role of a target model, which we attempt to recover from simulated TIMFs under different scenarios. In particular, we explore the effect of truncation of the spherical-harmonic expansion of the TIMF. These experiments are of practical importance since the truncation degree of recent data products is around 30 for M_2 tide and even lower for N_2 and O_1 tides. We also investigate how the amplitude of random noise in the spherical-harmonic series of TIMF affects the recovered conductivity structures. The theoretical basis of the forward and inverse problems is explained in Section 2. The inverse problem is solved effectively using the adjoint approach, the key formulas are derived in Appendix A. The synthetic inverse problems are constructed according to Section 3 and the results are presented in Section 4.

2 THEORY

2.1 Forward problem

The magnetic field $\mathbf{B}(\mathbf{r}; \omega_l)$ induced in the spherical Earth G with 3-D distribution of electrical conductivity $\sigma(\mathbf{r})$, comprising the oceans, the solid lithosphere and mantle and the highly conductive non-moving core, by a particular ocean tidal constituent with angular frequency ω_l and velocity distribution $\mathbf{u}(\mathbf{r}; \omega_l)$, is governed by the EMI equation (Velínský *et al.* 2018)

$$\text{curl} \left[\frac{1}{\sigma(\mathbf{r})} \text{curl} \mathbf{B}(\mathbf{r}; \omega_l) \right] - i\omega_l \mu_0 \mathbf{B}(\mathbf{r}; \omega_l) = \mu_0 \text{curl} [\mathbf{u}(\mathbf{r}; \omega_l) \times \mathbf{B}_M(\mathbf{r})]. \quad (1)$$

Here $\mathbf{r} = (r, \Omega) = (r, \vartheta, \varphi)$ is the position vector given by the radial distance r , colatitude ϑ and longitude φ , μ_0 is the magnetic permeability, $\mathbf{B}_M(\mathbf{r})$ denotes the main field, which is stationary and irrotational, and $\exp(-i\omega_l t)$ time dependence is assumed. In the surrounding insulating atmosphere, the magnetic field is described by a scalar magnetic potential

$$U(\mathbf{r}; \omega_l) = a \sum_{j=1}^{j_{\max}} \sum_{m=-j}^j \left[G_{jm}^{(e)}(\omega_l) \left(\frac{r}{a} \right)^j + G_{jm}^{(i)}(\omega_l) \left(\frac{a}{r} \right)^{j+1} \right] Y_{jm}(\Omega), \quad (2)$$

where $Y_{jm}(\Omega) = P_{jm}(\cos \vartheta) \exp(im\varphi)$ are the fully normalized complex spherical harmonic functions (Varshalovich *et al.* 1989), and $G_{jm}^{(e)}(\omega_l)$, $G_{jm}^{(i)}(\omega_l)$ are the complex spherical harmonic coefficients describing, respectively, the external field, originating in the magnetosphere and ionosphere and the internal, induced contribution.

At the Earth's surface $r = a$, the magnetic field is continuous,

$$\mathbf{B}(a, \Omega; \omega_l) = -\text{grad}|_{r=a} U(r, \Omega; \omega_l). \quad (3)$$

This boundary condition implies that the toroidal component of TIMF vanishes at $r = a$, since it can not be maintained in the insulating atmosphere. Moreover, in this particular setup, we do not deal with the induction by external sources and set $G_{jm}^{(e)}(\omega_l) = 0$ for all j, m . Thus, the EMI equation for a given motional forcing $\mathbf{u} \times \mathbf{B}_M$ and conductivity model σ predicts the induced field coefficients $G_{jm}^{(i)}(\omega_l)$ up to a pre-defined maximum truncation spherical harmonic degree $j_{\max}$.

2.2 Inverse problem

In order to formulate the inverse problem, we first introduce a finite-dimensional discretization of the conductivity distribution $\sigma(\mathbf{r})$. Let $\mathbf{m} \in \mathbb{R}^M$ and $\mathbf{m}_0 \in \mathbb{R}^{M_0}$ be real vectors of respective dimensions M and M_0 . Let $\xi_\alpha(\mathbf{r})$, $\alpha = 1, 2, \dots, M + M_0$ be piecewise constant functions describing the division of the entire sphere G into $M + M_0$ disjunctive blocks I_α of arbitrary size,

$$\xi_\alpha(\mathbf{r}) = \begin{cases} 1 & \mathbf{r} \in I_\alpha \\ 0 & \mathbf{r} \notin I_\alpha \end{cases}, \quad (4)$$

$$I_\alpha \cap I_{\alpha'} = \emptyset \quad \text{for } \alpha \neq \alpha', \quad (5)$$

$$\bigcup_{\alpha=1}^{M+M_0} I_\alpha = G. \quad (6)$$

Then, we can write

$$\log_{10} \frac{\sigma(\mathbf{m}, \mathbf{m}_0; \mathbf{r})}{\sigma_0} = \sum_{\alpha=1}^M m_\alpha \xi_\alpha(\mathbf{r}) + \sum_{\alpha=M+1}^{M+M_0} m_{0\alpha} \xi_\alpha(\mathbf{r}), \quad (7)$$

where $\sigma_0 = 1 \text{ S m}^{-1}$ is used for scaling. This general approach allows for the separation of regions, where the conductivity is sought for by the inversion (i.e. in the suboceanic upper mantle), and where the conductivity is represented by a fixed *a priori* value of $m_{0\alpha}$ (in the core, lower mantle, subcontinental upper mantle, lithosphere, sediment layers and oceans). As these are constant throughout our calculations, the dependency on $\mathbf{m}_0$ is dropped henceforth from the notation.

As shown in the previous section, the forward problem calculates for a given tidal constituent of frequency ω_l and given conductivity distribution $\sigma(\mathbf{m}; \mathbf{r})$ a set of internal field coefficients $G_{jm}^{(i)}(\mathbf{m}; \omega_l)$. The comprehensive inversion (Sabaka *et al.* 2020) or the dedicated residual inversion by Grayver & Olsen (2019) provide the sets of coefficients $G_{jm}^{(i, \text{obs})}(\omega_l)$ (except for different normalization) derived from the satellite data. In synthetic tests, the coefficients $G_{jm}^{(i, \text{obs})}(\omega_l)$ are provided directly by a forward calculation using a known target conductivity model. A natural definition of misfit between the predicted and observed fields can be formulated as follows:

$$\chi^2(\mathbf{m}) = \sum_l \sum_{jm} \sum_{j'm'} N_{jj'} \left(\frac{a}{r_s} \right)^{\frac{l+j'}{2}} \left[\overline{G_{jm}^{(i)}(\mathbf{m}; \omega_l)} - \overline{G_{jm}^{(i, \text{obs})}(\omega_l)} \right] W_{jmj'm'}^l \left[G_{j'm'}^{(i)}(\mathbf{m}; \omega_l) - G_{j'm'}^{(i, \text{obs})}(\omega_l) \right]. \quad (8)$$

Here, the bar denotes a complex conjugation, the normalization factor is defined as

$$N_{jj'} = \frac{1}{8\pi} \sqrt{(2j+1)(j+1)(2j'+1)(j'+1)}, \quad (9)$$

and the weighting matrix must satisfy

$$\overline{W_{j'm'jm}^l} = W_{jmj'm'}^l \quad (10)$$

since misfit is a real-valued function. The summation over l spans single or multiple tidal constituents $l = (M_2, N_2, O_1)$, and the summations over spherical harmonic degrees j and j' are truncated at j_{maxD}^l defined by the respective data set. The spherical harmonic orders m and m' range from $-j$ to j , and $-j'$ to j' , respectively. The weighting matrix $W_{jmj'm'}^l$ is the inverse covariance matrix, ideally provided for the individual sets of $G_{jm}^{(i, \text{obs})}(\omega_l)$ from data analysis. In a simplified case, it takes a diagonal form with standard errors of individual coefficients s_{jm}^l ,

$$W_{jmj'm'}^l = \delta_{jj'} \delta_{mm'} \frac{1}{(s_{jm}^l)^2}, \quad (11)$$

or, in the most trivial case, the form of a unit matrix multiplied by a scalar weight w^l ,

$$W_{jmj'm'}^l = \delta_{jj'} \delta_{mm'} \frac{1}{(s^l)^2} = \delta_{jj'} \delta_{mm'} w^l. \quad (12)$$

In the latter case, one can show, by expressing the gradient of the internal part of the magnetic potential (2) in vector spherical harmonics (Varshalovich *et al.* 1989), that the misfit defined by eq. (8) corresponds to the L_2 norm of the differences between the predicted and observed magnetic fields over sphere S of radius $r_s \geq a$,

$$\chi^2(\mathbf{m}) = \frac{1}{2S} \sum_l \int_S \left| \frac{\mathbf{B}(\mathbf{m}; r_s, \Omega; \omega_l) - \mathbf{B}^{\text{obs}}(r_s, \Omega; \omega_l)}{s^l} \right|^2 dS. \quad (13)$$

We assume the satellite radial distance $r_s = 6771.2 \text{ km}$, corresponding to the altitude of 400 km throughout the paper.

We are now ready to define the solution of regularized inverse problem as

$$\hat{\mathbf{m}}(\lambda) = \arg \min_{\mathbf{m} \in \mathbb{R}^M} [\chi^2(\mathbf{m}) + \lambda R^2(\mathbf{m})], \quad (14)$$

where $R^2(\mathbf{m})$ is the regularization functional and λ is the regularization parameter balancing the data misfit and regularization.

In this study, we use the regularization term in the form

$$R^2(\mathbf{m}) = \frac{1}{a} \int_G |\text{grad}_H \log_{10} \sigma(\mathbf{m}; \mathbf{r})|^2 dV, \quad (15)$$

where grad_H is the 2-D horizontal gradient operator,

$$\text{grad}_H = \mathbf{e}_\varphi \frac{1}{r \sin \vartheta} \frac{\partial}{\partial \varphi} + \mathbf{e}_\vartheta \frac{1}{r} \frac{\partial}{\partial \vartheta}. \quad (16)$$

Since the conductivity is expected to be a smooth function of horizontal coordinates, the regularization term R^2 should suppress sharp extrema and noisy structure by limiting the first horizontal derivatives in each layer. We integrate over the spherical Earth G in eq. (15), but the model layers with a 1-D conductivity structure can be omitted since the integrand is equal to zero there.

The total 3-D gradient operator can also be used in order to suppress the variations in both horizontal and vertical directions. We found additional regularization in the radial direction unnecessary in the presented computations since the recovered models are sufficiently smooth in the radial direction.

The solution of a regularized inverse problem (14) is obtained by the limited-memory quasi-Newton minimization approach with the Broyden–Fletcher–Goldfarb–Shanno formula (Press *et al.* 1992, section 10.7) used for gradual updates of the approximate inverse of the Hessian. The data misfit derivative is calculated efficiently using the adjoint approach as explained in Appendix A. If regularization is applied, we run a series of regularized inverse problems. The first inverse problem is started with a 1-D background mantle conductivity profile and each subsequent inverse problem is started with the solution of previous minimization. The regularization parameter λ is decreasing from the first to the last inverse problem. By choosing the optimal value of $\hat{\lambda}$ near the maximum inflexion point of the L-curve (Hansen 1992), we obtain the corresponding solution $\hat{\mathbf{m}}(\hat{\lambda})$.

3 NUMERICAL SETUP

3.1 Target conductivity model

We represent the target conductivity model by a sphere with 203 layers. The finest resolution is used in the lithosphere and in the oceans. There are 140 layers in the upper 21 km in order to capture the heterogeneous 3-D structure of near-surface conductivity and tidal forcing. In total, 60 layers discretize the Earth's mantle: 20 layers in the upper mantle down to 406 km depth, 16 layers in the transition zone and 24 layers in the lower mantle from 676 km down to CMB at 2876 km. The remaining three layers discretize the Earth's core.

We set up the conductivity structure in our model as follows. In the oceans, the annual mean values of ocean conductivity from collocated temperature and salinity measurements assembled by Tyler *et al.* (2017) are used. The largest conductivity values as well as the largest lateral variations are located from the surface down to approximately 1 km depth. The spatial average conductivity of seawater is $3.31 \pm 0.23 \text{ S m}^{-1}$. The ocean floor is modelled with realistic TPX09 bathymetry. Below the seafloor or below the land surface, layers of continental, coastal and two types of open ocean sediments of various thicknesses with respective conductivities of 0.03, 0.5, 0.8 S m^{-1} (shallow) and 0.02 S m^{-1} (deep) are embedded using the approach by Everett *et al.* (2003). A conductivity of $3.48 \times 10^{-4} \text{ S m}^{-1}$ is assigned to igneous rocks in the crust (Grayver *et al.* 2017).

The 1-D and 3-D upper mantle conductivities are considered above and below the 143 km depth, respectively. The conductivity values are assigned according to the 3-D conductivity model WINTERC-e (Martinec *et al.* 2021). We point out that no EM data were used in the creation of WINTERC-e, providing a completely independent information. WINTERC-e is a derivative of the WINTERC-G model (Fullea *et al.* 2021), which is constructed using the LitMod software inverting seismic waveform tomography, global satellite gravity, surface elevation and heat flow data. WINTERC-e is constructed by combining the WINTERC-G's thermal and compositional fields with the results of laboratory experiments on the conductivity of upper mantle minerals and melt. The WINTERC-e model provides two end-members: Wd-emin and Wd-emax. The Wd-emin model represents the least conductive model derived under dry mantle conditions. The presence of water and hydrous melt is considered in Wd-emax. Since the presence of water and melt increases the conductivity of the rock aggregate, the Wd-emin model can be viewed as the lower limit for the upper-mantle conductivity. Besides that, Wd-emin is less affected by the experimental uncertainties of proton conduction in upper mantle minerals, composition of the melt, water partition coefficients between melt/solid, selection of water solubility model and behaviour of melt in a peridotitic medium. However, based on the results by Martinec *et al.* (2021), Wd-emax will provide a more suitable prior for future inversion of satellite-observed TIME. Therefore we use the Wd-emax model as the testbed in this paper. The 1-D globally averaged Wd-emax model is used as a background model.

WINTERC-e provides the conductivity structure from the Moho discontinuity down to 402 km. The model contains 201 layers, each layer is 2 km thick. The conductivity structure is laterally discretized on a triangular grid with an average 225 km lateral interknot spacing. We use a coarse-resolution version of the WINTERC-e model with every 10 layers merged into one layer, and lateral averaging to the computational grid of the forward solver. The resulting target model is depicted at the depth of 153, 233, 313 and 395 km in the left-hand column of Fig. 3. In the upper mantle regions above 143 km depth, we use a 1-D conductivity profile computed as a global average of the WINTERC-e model. In order to avoid numerical instabilities of the forward solver, we enforce a lower conductivity bound of 10^{-4} S m^{-1} .

The conductivity in the transition zone and lower mantle is 1-D. We use the 1-D conductivity profile of Grayver *et al.* (2017). The Earth's core has a constant conductivity of $2 \times 10^5 \text{ S m}^{-1}$ according to Velínský (2013). The conductivity of both lower mantle and core has little impact on the TIMF since the penetration depth of tidal signals is much shallower.

3.2 Motional forcing

Tidal circulation interacts with the main magnetic field $\mathbf{B}_M$ and generates the motional forcing, $\mathbf{u} \times \mathbf{B}_M$, which stands on the right-hand side of the EMI eq. (1). We use the tidal circulation from the TPX09 model (Egbert & Erofeeva 2002, updated), which is a state-of-the-art hydrodynamic tidal model constrained by satellite altimetry assimilation. The TPX09 model is a barotropic model that is based on the shallow water equations. The ocean velocities in barotropic models are assumed to be constant in the entire water column. Therefore, local bathymetry $b(\vartheta, \varphi)$ is used to scale the vertically integrated velocity $\mathbf{U}(\vartheta, \varphi; \omega_l)$ provided by TPX09 into the velocity,

$$\mathbf{u}(r, \vartheta, \varphi; \omega_l) = \begin{cases} \mathbf{U}(\vartheta, \varphi; \omega_l)/b(\vartheta, \varphi) & \text{for } r \geq a - b(\vartheta, \varphi), \\ \mathbf{0} & \text{otherwise.} \end{cases} \quad (17)$$

We use the 13th generation International Geomagnetic Reference Field IGRF-13(2015.0) model (Alken *et al.* 2021) in the geodetic coordinates for $\mathbf{B}_M$. Its three vector components are calculated in every point of the computational grid up to degree 13. In detail, we use the IGRF-13 Schmidt semi-normalized spherical harmonic coefficients up to degree 13 for year 2015. The grid point values of the $\mathbf{B}_M$ are evaluated using the geodetic coordinates. The MMF is 3-D in our computations, that is we consider not only the lateral but also the relatively small vertical variations of the MFF.

3.3 Forward solver ElmgFD

We use the EMI solver ElmgFD (Velínský *et al.* 2018) to compute the TIMF in the forward and inverse problems. ElmgFD is based on the weak formulation of the EMI equation. It uses the complex vector spherical harmonic functions in angular coordinates and piecewise-linear 1-D finite elements in radial coordinate to numerically discretize the governing equations. The computation of TIMF then reduces to finding the solution $\mathbf{x}$ of the system of linear equations,

$$\mathbf{Ax} = \mathbf{b}, \quad (18)$$

where the right-hand side $\mathbf{b}$ is composed of source integrals. The system matrix $\mathbf{A}$ has block-tridiagonal structure due to radial finite-element discretization. If the conductivity is 1-D, the blocks are banded matrices with two sub- and superdiagonals due to decoupling in spherical harmonics. In that case, the factorization is fast and the problem is solved in one iteration. In the general 3-D case, each block is a dense matrix. We search for the solution of eq. (18) using the BiCGStab(2) iterative method (Sleijpen & Fokkema 1993). The stopping criterion limits the relative residual error ϵ ,

$$\epsilon = \frac{\|\mathbf{Ax} - \mathbf{b}\|}{\|\mathbf{Ax}_0 - \mathbf{b}\|}, \quad (19)$$

where $\mathbf{x}_0$ is the initial guess of the solution. We use $\epsilon \leq 10^{-6}$, the effect of more restrictive stopping criterion on the solution is negligible. The pre-conditioner for the iterative solution is the system matrix of 1-D conductivity profile. The matrix-vector product is computed effectively using the properties of spherical harmonics and the fast Fourier transform. Note that the procedure does not require to store the full system matrix. The numerical implementation uses the OpenMP shared-memory parallelization across spherical-harmonic degree, order and layer.

3.4 Synthetic TIMFs

Given the target conductivity model WINTERC-e Wdmax, and the TPX09 tidal forcing, we calculate the induced spherical-harmonic coefficients $G_{jm}^{(i, \text{obs})}(\omega_l)$ for tidal constituents $l = (M_2, N_2, O_1)$, using the ElmgFD solver with a 0.25° computational grid and a maximum spherical harmonic degree $j_{\text{max}} = 480$ (from now on referenced as high-resolution, HR).

Fig. 1 shows the Lowes–Mauersberger spectrum of the M_2 TIMF for the HR target model for observable degrees 1–18 at the Earth's surface, together with the spectra of three recent satellite-derived data sets: GO19 (Grayver & Olsen 2019), MTL601 and MTL701 (Sabaka *et al.* 2020, released as Swarm Level 2 products by ESA, <https://swarm-diss.eo.esa.int/#swarm%2FLevel2longterm%2FMTI>). Except for degree 1, there is good agreement between the spectra of the HR target model, and the GO19 data set. That confirms the suitability of our choice of the target model for synthetic inversion tests. We refer to Martinec *et al.* (2021) for a more detailed analysis.

3.5 Lateral resolution of forward modelling

The HR solution of the forward EMI problem is computationally demanding. For a single tidal constituent it takes over 50 hr on a recent 8-core PC to reach the convergence of the BiCGStab(2) iterations. The inverse EMI problem then requires hundreds of forward runs for each

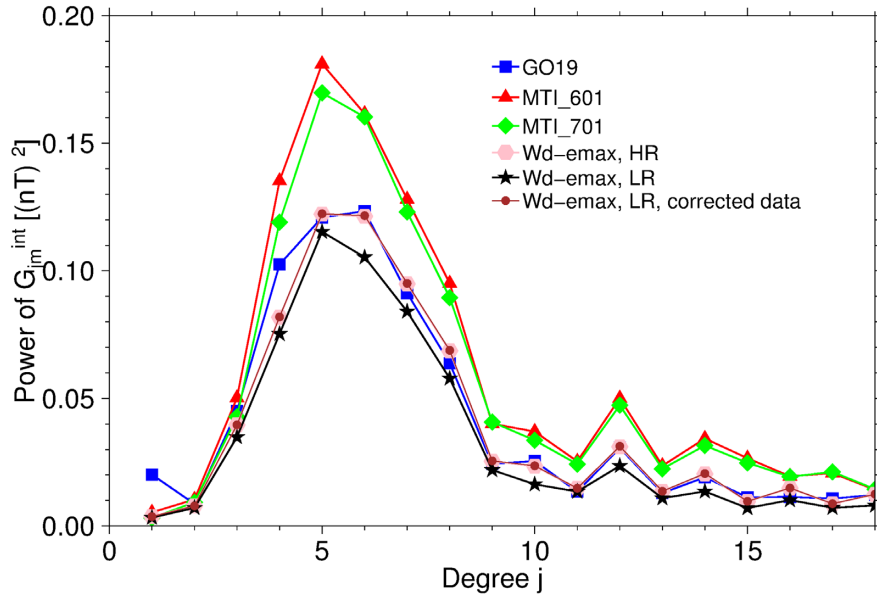


Figure 1. Lowes–Mauersberger spectra of the M_2 TIMF of the GO19, MTL601 and MTL701 satellite-derived data sets together with the spectra computed using the high and low resolutions and the low resolution with the data correction.

test setup, making the use of the HR solution unfeasible. Reducing the truncation degree of the forward solver to $j_{\max} = 60$, corresponding to $2^\circ \times 2^\circ$ lateral resolution (referred to as low-resolution, LR, henceforth), yields acceptable computational times for each forward or adjoint problem solution in the order of minutes.

However, as Fig. 1 clearly demonstrates, reducing the spatial resolution has a significant impact on the calculated TIMF even at degrees below 18. Attempting an inversion of HR synthetic data with LR forward modelling yields a poor correlation between the inverted and target models, as discussed in the Section 4.1 (test case Md60p0nc).

In order to bypass this problem, we propose the following linearized correction scheme. It is based on the assumption, that the short-wavelength EM field on degrees $j > 60$ are primarily governed by the motional forcing and local variations of bathymetry, seawater and sediment conductivity. They are only weakly sensitive to the conductivity of the structures beneath 150 km depth in the upper mantle. The first-order effect of shorter wavelength can be removed from synthetic data by using the corrected data coefficients

$$\widetilde{G}_{jm}^{(i, \text{obs})}(\omega_l) = G_{jm}^{(i, \text{obs})}(\omega_l) - \left[G_{jm}^{(i, \text{HR})}(\omega_l) - G_{jm}^{(i, \text{LR})}(\omega_l) \right], \quad (20)$$

where $G_{jm}^{(i, \text{HR})}(\omega_l)$ and $G_{jm}^{(i, \text{LR})}(\omega_l)$ are the induced field coefficients computed in the background 1-D model using the high and low resolutions, respectively. The effect of the linear correction on the observable TIMF spectra is also shown in Fig. 1.

Thus, the HR solution of the forward problem is needed only once, and remains outside of the inverse-problem iteration loop. In the future inversion of satellite-observed TIMFs, the correction could be also applied iteratively with successive updates of the 3-D conductivity model from the inversion. The number of HR runs of the forward problem still remains in the order of units, rather than hundreds.

3.6 Synthetic test setup

All inversions of the synthetic data set use a common basic setup.

We consider satellite data at 400 km altitude and 1-D subcontinental upper mantle conductivity structure in the inverse problem. This configuration is relevant for the inversion of real data. It could be used to recover the conductivity structure from the Swarm satellite measurements since the Swarm satellites are orbiting between 400 and 550 km altitude. Note that the use of 1-D instead of 3-D conductivity beneath continents is not a limiting factor since the resolution power of the tidal signals below the continents is very low.

We use single tide M_2 and $w^l = 1 \text{ (nT)}^{-2}$ in (12) which corresponds to the unit weighting matrix in all tests except for the tests in Section 4.4. In Section 4.4, we use also N_2 and O_1 tides and we experiment with larger weights for these additional tidal constituents in the combined $M_2 + N_2 + O_1$ configurations.

We search for the decimal logarithm of conductivity of M disjunctive suboceanic blocks in $N_l = 13$ layers in the 143–406 km depth range. Each layer is 20 km thick with the exception of the deepest layer, which is 23 km thick. The size of each block depends on the truncation degree $j_{\max D}$ of input tidal signal. The truncation degree $j_{\max D}$ corresponds to the grid interval $\Delta = (120/j_{\max D})^\circ$ (Martinec 1989). We set the size of each block to $(2\Delta) \times (2\Delta)$, that is we use 4° , 6° , 8° and 10° blocks if $j_{\max D}$ is equal to 60, 40, 30 and 24, respectively. In each layer, the suboceanic upper mantle blocks are spatially distributed according to the mask $\mathcal{M}_a$ which takes values 0 or 1 (Figs 2a–d). The block I_α is suboceanic if its mask value is equal to 1, $\mathcal{M}_a(I_\alpha) = 1$. We use the same mask $\mathcal{M}_a$ in each layer where we search for the 3-D conductivity

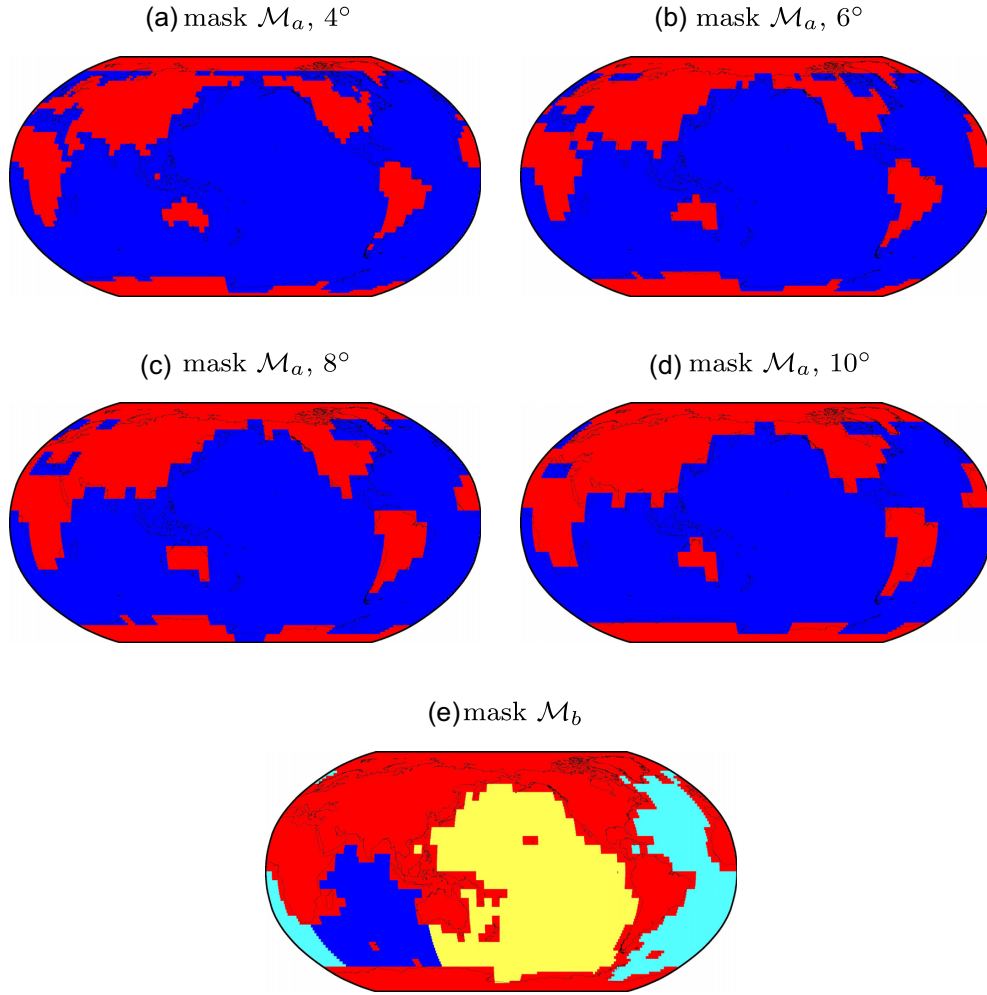


Figure 2. Masks used in the paper. The appropriate mask $\mathcal{M}_a$ is applied to the inverse problem and mask $\mathcal{M}_b$ is applied to the computation of the correlation coefficient (21). In mask $\mathcal{M}_b$, the subsets corresponding to the Pacific, Atlantic and Indian Oceans are plotted in yellow, light blue and dark blue, respectively.

structure. Outside of the mask $\mathcal{M}_a$, the electrical conductivity is fixed to an *a priori* value corresponding to the 1-D background model. The total number of model parameters is $M = (2710, 1143, 618, 381) \times N_l$ if $j_{\max D}$ is equal to 60, 40, 30 and 24, respectively.

The mask $\mathcal{M}_a$ is constructed from the 2° ocean-land mask. The respective criterion to set mask value of the block I_α to $\mathcal{M}_a(I_\alpha) = 1$ in the $4^\circ, 6^\circ, 8^\circ$ and 10° $\mathcal{M}_a$ mask is at least one, four, nine and sixteen sub-oceanic grid points within the block I_α , otherwise $\mathcal{M}_a(I_\alpha) = 0$. The criterion is chosen in order to eliminate sub-continental blocks where the inverse problem is insensitive to the conductivity structure. However, the criterion accepts blocks that are only partly beneath oceans since certain sensitivity is still present. Additionally, we exclude the Arctic Ocean from mask $\mathcal{M}_a$ for colatitudes $\vartheta < 25^\circ$. The reason is that the sensitivity of the inverse problem on the conductivity structure beneath the Arctic Ocean is poor. We do not introduce the Southern-Ocean mask since we consider the Pacific, Atlantic and Indian Oceans to extend to the Antarctica in the south.

3.7 Quality assessment

The quality of the recovered model is assessed by calculating the Pearson's correlation coefficient C_r between the suboceanic reconstructed and target log-conductivities,

$$C_r(\sigma, \sigma_T, r) = \frac{\int_{\Omega} \mathcal{M}_b(\Omega) \log_{10} [\sigma(r, \Omega) / \bar{\sigma}(r)] \log_{10} [\sigma_T(r, \Omega) / \bar{\sigma}_T(r)] d\Omega}{\sqrt{\left\{ \int_{\Omega} \mathcal{M}_b(\Omega) \log_{10}^2 [\sigma(r, \Omega) / \bar{\sigma}(r)] d\Omega \right\} \left\{ \int_{\Omega} \mathcal{M}_b(\Omega) \log_{10}^2 [\sigma_T(r, \Omega) / \bar{\sigma}_T(r)] d\Omega \right\}}}, \quad (21)$$

where σ is the reconstructed conductivity, σ_T is the target conductivity regridded to 2° grid and overbar denotes the geometric mean of values in blocks where $\mathcal{M}_b(I_\alpha) = 1$. The mask $\mathcal{M}_b$ is shown in Fig. 2(e). It is a 6° mask constructed from the 0.25° ocean-land mask by applying the criterion that $\mathcal{M}_b(I_\alpha) = 1$ if all grid cells within the block are suboceanic. The criterion is adjusted in a few blocks to allow using the mask $\mathcal{M}_b$ to assess the quality of the recovered model regardless of $j_{\max D}$ and $\mathcal{M}_a$ in our tests.

Table 1. Numerical tests that we performed in this study. The weights w^j are in $(\text{nT})^{-2}$.

Test	Correction	j_{maxD}	Noise	w^{M_2}	w^{N_2}	w^{O_1}	Average C_r
Md60p0nc	No	60	0%	1	0	0	0.09
Md60p0	Yes	60	0%	1	0	0	0.76
Md40p0	Yes	40	0%	1	0	0	0.75
Md30p0	Yes	30	0%	1	0	0	0.71
Md24p0	Yes	24	0%	1	0	0	0.69
Md60p5j31	Yes	60	5% for $j \geq 31$	1	0	0	0.75
Md60p5j21	Yes	60	5% for $j \geq 21$	1	0	0	0.72
Md60p5j11	Yes	60	5% for $j \geq 11$	1	0	0	0.56
Md60p1j1	Yes	60	1% for $j \geq 1$	1	0	0	0.60
Md60p3j1	Yes	60	3% for $j \geq 1$	1	0	0	0.28
Md60p5j1	Yes	60	5% for $j \geq 1$	1	0	0	0.17
Nd60p0	Yes	60	0%	0	1	0	0.66
Od60p0	Yes	60	0%	0	0	1	0.73
MNOd60p0	Yes	60	0%	1	1	1	0.77
MNOd60p0w2	Yes	60	0%	1	2	2	0.75
MNOd60p0w4	Yes	60	0%	1	4	4	0.72

4 RESULTS

We perform a series of numerical tests summarized in Table 1. We test how the truncation degree of the input tidal signal, noise in the input tidal signal and additional tidal constituents affect the quality of the conductivity reconstruction.

4.1 Noise-free synthetic data with truncation degree $j_{\text{maxD}} = 60$

In the first two tests, Md60p0nc and Md60p0, we make two important simplifications. First, synthetic data are noise-free. Secondly, the truncation degree j_{maxD} of spherical-harmonic expansion of input data corresponds to the maximum spherical harmonic degree $j_{\text{max}} = 60$ used in the EMI solver in the inverse problem.

We apply the regularization as explained in Section 4.3. In the absence of noise, the regularization is not critically needed but the reconstructed conductivity is closer to the target model if regularization is used.

First of all, we demonstrate the importance of corrected data coefficients $\widetilde{G}_{jm}^{(i,\text{obs})}$. We use original data coefficients $G_{jm}^{(i,\text{obs})}(\omega_l)$ in the Md60p0nc test and corrected data coefficients $\widetilde{G}_{jm}^{(i,\text{obs})}$ in the Md60p0 test. The respective averages of correlation coefficients (21) in the Md60p0nc and Md60p0 tests beneath the Global Ocean (according to the $\mathcal{M}_b$ mask) are 0.09 and 0.76. Thus, the conductivity reconstructed in the Md60p0 test is much closer to the target conductivity than the conductivity reconstructed in the Md60p0nc test. We discuss the Md60p0 test in the rest of this section.

The conductivity model recovered in the Md60p0 test is depicted in the middle column of Fig. 3. The target model and the differences in absolute value between the target model regridded to the 2° grid and the recovered model are depicted in the left and right-hand columns of Fig. 3, respectively. The rows correspond to the model layers number 1, 5, 9 and 13 which are centred at the depths of 153, 233, 313 and 395 km, respectively. The subcontinental regions are shaded in the recovered model since we do not search for the conductivity structure beneath continents. If we compare the target and recovered models in Fig. 3, we can make a subjective observation that the characteristic conductivity structure of the target model has been recovered successfully. However, the match is not perfect. For example, the South-Pacific conductivity anomaly in the recovered model is overly strong in the upper-most layer but it is too weak in the other layers.

The correlation coefficients computed in the Md60p0 test are depicted in Fig. 4, see the blue profile. The dashed lines denote the 95 per cent confidence interval of the Pearson's correlation coefficient. Panels (a), (b), (c) and (d) correspond to the correlation coefficients computed beneath the Global, Pacific, Atlantic and Indian Oceans, respectively.

The correlation coefficient beneath the Global Ocean is greater than 0.8 from the first layer at 153 km depth down to the seventh layer at 273 km depth. The confidence interval is rather narrow. It implies that the global conductivity structure has been recovered to certain extent. That is in accordance with our finding based on the visual inspection of Fig. 3.

In all panels in Fig. 4, the correlation coefficient first increases with depth, it reaches its maximum at certain depth and then it decreases. The maximum of the correlation coefficient is reached either in the fifth layer at 233 km depth beneath the Global, Pacific and Atlantic Oceans or in the sixth layer at 253 km depth beneath the Indian Ocean. In the deep layers, the confidence interval is wider and the recovered conductivity is more distant from the target conductivity. For example, the correlation coefficient in the deepest layer at 395 km depth is 0.49 beneath the Global Ocean.

Additionally, the correlation profile varies regionally. In layers deeper than 280 km, the correlation coefficient decreases with depth more rapidly beneath the Pacific and Atlantic Oceans than beneath the Indian Ocean. The best agreement between the target and reconstructed conductivity structures is achieved beneath the Indian Ocean. We argue that the reason is the strong M_2 signal beneath the Indian Ocean

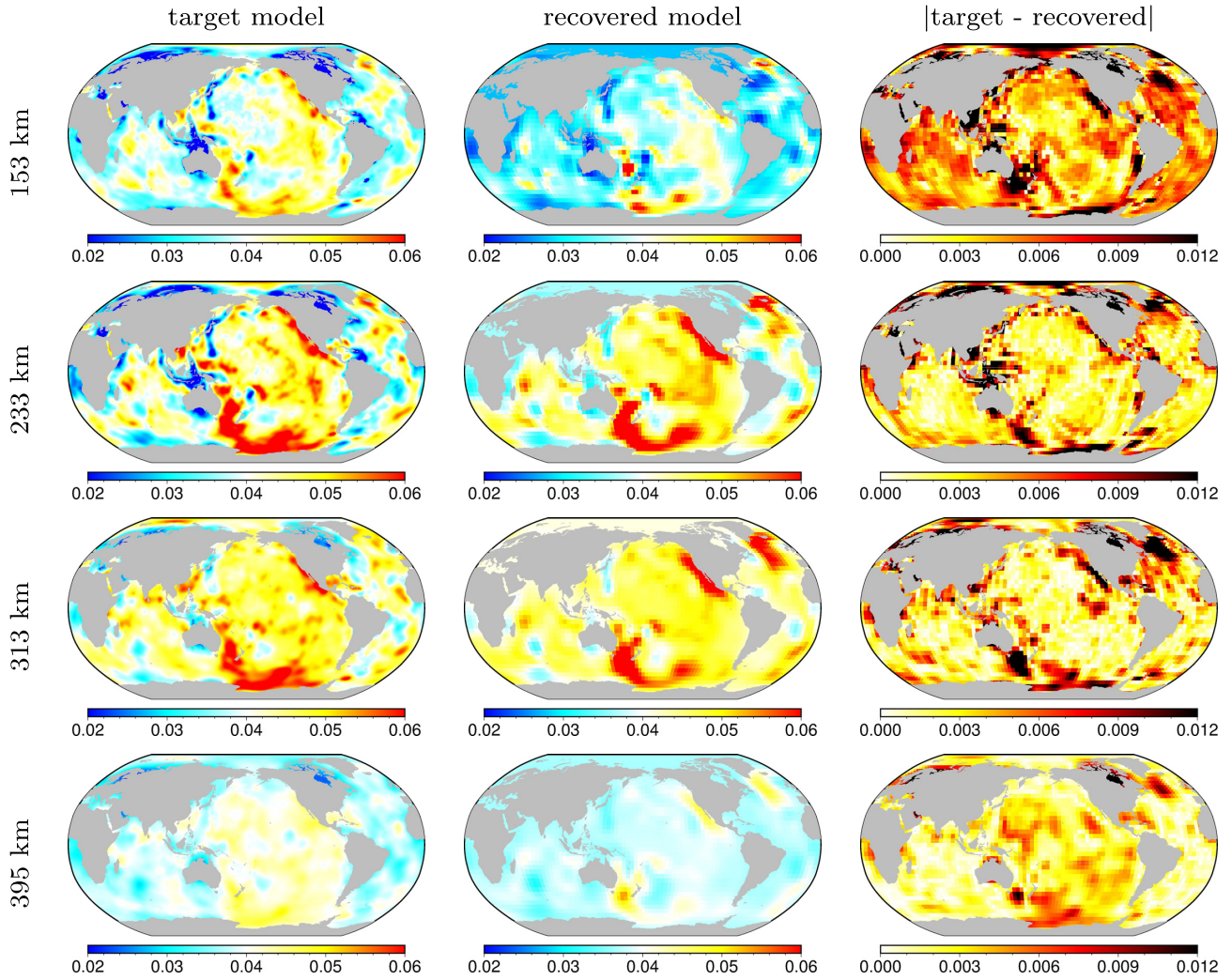


Figure 3. Target model (left-hand column), model recovered in the Md60p0 test (middle column) and the differences in absolute value (right-hand column) in S m^{-1} at the depths of 153 km (top row), 233, 313 and 395 km (bottom row). The conductivity structure beneath continents is 3-D in the target model but it is 1-D in the inverse problem.

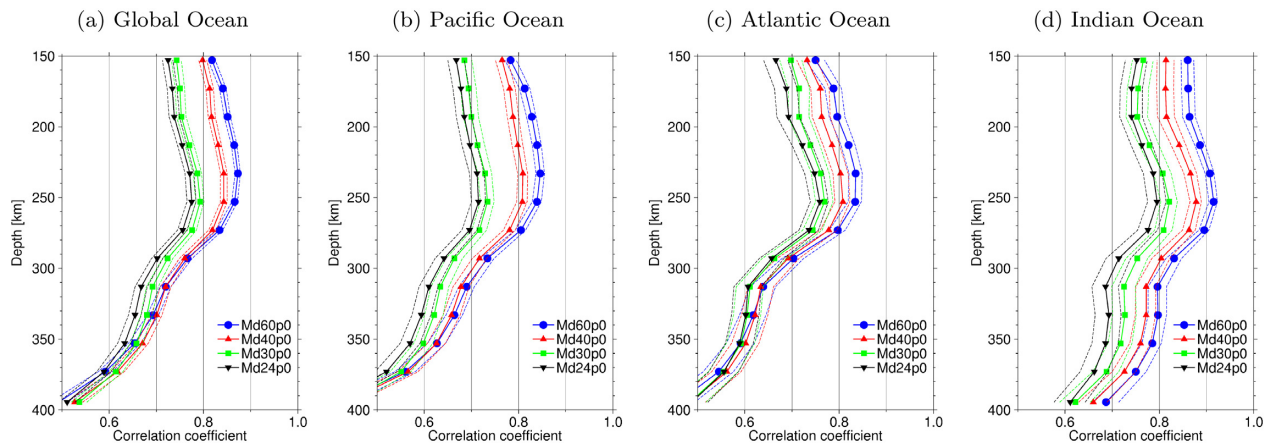


Figure 4. Impact of the truncation degree of the spherical-harmonic expansion of input tidal signal on the depth profiles of correlation coefficient beneath the (a) Global Ocean, (b) Pacific Ocean, (c) Atlantic Ocean and (d) Indian Ocean in the Md60p0, Md40p0, Md30p0 and Md24p0 tests.

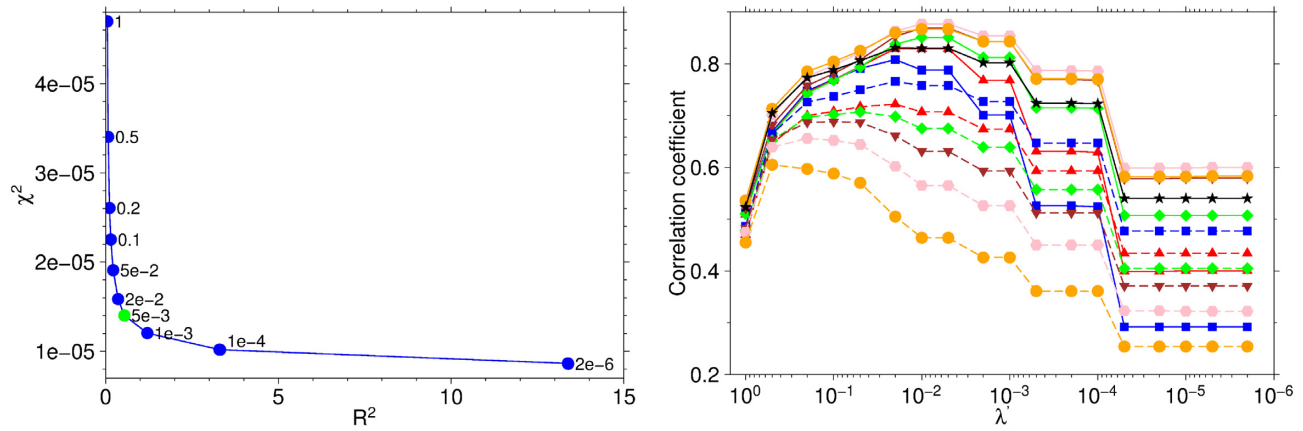


Figure 5. *L*-curve (left-hand panel) and correlation coefficients (right-hand panel) computed in the Md60p5j31 test. Left-hand panel: *L*-curve shows the data misfit $\chi^2(\mathbf{m})$ with respect to the regularization term $R^2(\mathbf{m})$. The normalized regularization parameter λ' is plotted at each point. The optimal result corresponds to the green point. Right-hand panel: correlation coefficient plotted with respect to the normalized regularization parameter λ' for all layers. The solid lines with blue squares, red triangles, green diamonds, brown inverted triangles, pink hexagons, orange circles and black stars correspond to the seven upper-most layers from top to bottom, respectively. The dashed lines correspond to the six deeper layers, the colours and symbols are repeated in the same order.

(fig. 2 in Grayver & Olsen 2019). The respective average correlation coefficients beneath the Pacific, Atlantic and Indian Oceans are 0.727, 0.706 and 0.833.

4.2 Noise-free synthetic data with limited truncation degree $j_{\max D} < 60$

The truncation degree $j_{\max D} = 60$ of input data is higher than that of recent TIMF data sets. Grayver & Olsen (2019) and Sabaka *et al.* (2020) extracted the TIMF of M_2 tidal constituent using truncation degrees of 28 and 36, respectively. Consequently, the sensitivity of the inverse problem on the truncation degree of input data is a question of practical matter.

Besides $j_{\max D} = 60$ in the reference Md60p0 test, we inspect three truncation degrees $j_{\max D} = 40, 30$ and 24 in the tests Md40p0, Md30p0 and Md24p0, respectively. The corresponding correlation profiles are included in Fig. 4. The differences between the individual correlation profiles are larger in the upper layers which is caused by the more complex conductivity structure. The agreement between the target and reconstructed conductivity structures is gradually decreasing with the decreasing $j_{\max D}$. The average Md40p0, Md30p0 and Md24p0 Global-Ocean correlation coefficients are 0.75, 0.71 and 0.69, respectively. Nevertheless, in all these profiles, the Global-Ocean correlation coefficients are larger than 0.7 in the upper eight layers in the 153–293 km depth range. We regard this result as encouraging for the reconstruction of 3-D upper mantle conductivity structure from real data. The truncation degrees of modern TIMF data products approximately correspond to the Md24p0 and Md30p0 configurations.

4.3 Noisy synthetic data with truncation degree $j_{\max D} = 60$

We now perform several tests with noise added to synthetic TIMF in order to assess how noise affects the accuracy of the recovered model. We consider synthetic TIMF with the truncation degree $j_{\max D} = 60$ in all tests with added noise. We consider six configurations. In configurations Md60p1j1, Md60p3j1 and Md60p5j1, we add 1, 3 and 5 per cent Gaussian noise to the spherical harmonic coefficients of all degrees, respectively. In configurations Md60p5j31, Md60p5j21 and Md60p5j11, we add 5 per cent Gaussian noise to the spherical harmonic coefficients of degrees greater or equal than 31, 21 and 11, respectively.

The presence of noise requires regularization. We solve the chain of regularized inverse problems (14) with the regularization term (15) and gradually decreasing regularization parameter λ . There are 18 inverse problems in the chain. The regularization parameter in the first problem is $\lambda_1 = 6.371 \times 10^{-4}$ (an arbitrary choice related to the normalization of the regularization term). The successive normalized regularization parameters $\lambda' = \lambda/\lambda_1$ are then equal to 1, 0.5, 0.2, 0.1, ..., 2×10^{-6} . The optimal value $\tilde{\lambda}'$ of the normalized regularization parameter is chosen according to the *L*-curve. The visual inspection of the reconstructed conductivity structure is also useful. If the conductivity appears noisy, the regularization parameter should be increased. The procedure is demonstrated on the Md60p5j31 configuration.

The Md60p5j31 *L*-curve is depicted in the left-hand panel of Fig. 5. The points on the *L*-curve are labelled by the values of the regularization parameter. If several points overlap, we label only the point with the lowest λ' . We choose $\tilde{\lambda}' = 5 \times 10^{-3}$ as the optimal value of the normalized regularization parameter. Fig. 6 shows the target and reconstructed conductivity at 153, 253 and 395 km depths. The reconstructed conductivity is plotted for the largest, optimal and smallest values of the normalized regularization parameter. Apparently, the optimally regularized conductivity in the second column does not lack details, but at the same time, it is not noisy.

The advantage of the synthetic inverse problem is that we can verify our choice of the optimal regularization parameter by inspecting the correlation coefficients. The right panel of Fig. 5 shows the Global-Ocean correlation coefficients computed for all inspected layers

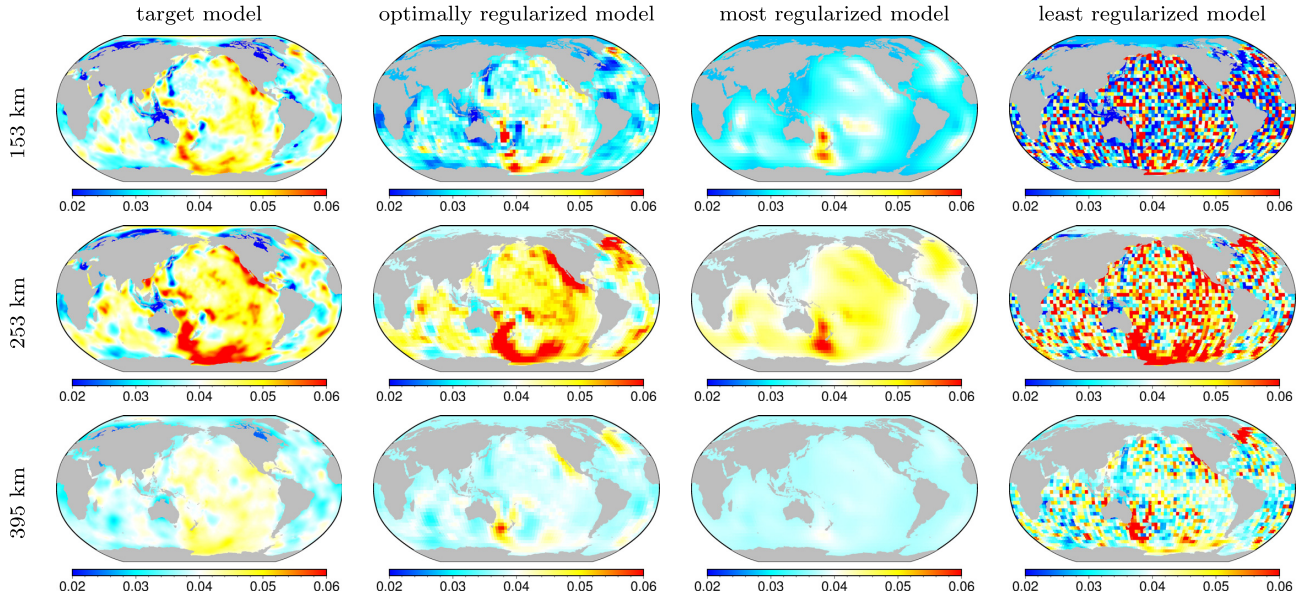


Figure 6. Target and reconstructed conductivity models in the Md60p5j31 test. The normalized regularization parameters λ' used in the most regularized, optimally and least regularized models are 1 , 5×10^{-3} and 2×10^{-6} , respectively.

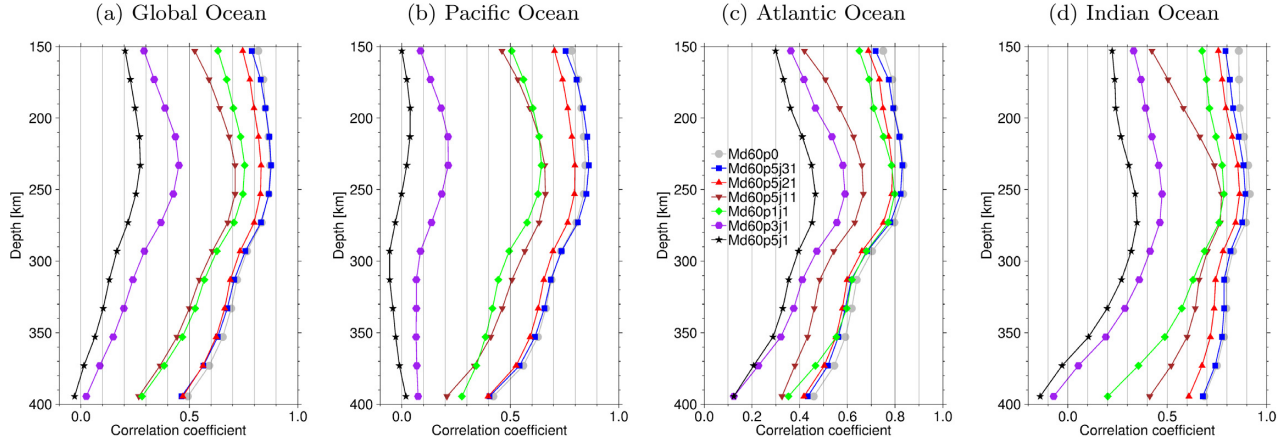


Figure 7. Impact of noise added to the particular spherical-harmonic degrees j of input tidal signal on the depth profiles of correlation coefficient beneath the (a) Global Ocean, (b) Pacific Ocean, (c) Atlantic Ocean and (d) Indian Ocean in the Md60p0, Md60p5j31, Md60p5j21, Md60p5j11, Md60p1j1, Md60p3j1 and Md60p5j1 tests. The colour coding denoted for the Atlantic Ocean is common for all panels.

and all values of the normalized regularization parameter. The dependency of the correlation coefficient on the normalized regularization parameter, $C_r(\lambda')$, is similar in each layer. At first, the correlation coefficient increases from its initial value $C_r(1)$ with decreasing normalized regularization parameter. The reason is the appearance of local conductivity structure due to relaxing the smoothness restraints. The correlation coefficient reaches its maximum for some value of the normalized regularization parameter. If we further decrease the normalized regularization parameter, the misfit (see the left-hand panel) decreases but it is at the cost of noise that appears in the reconstructed conductivity. The noise causes the decrease of the correlation coefficient. There is not a single value of the regularization parameter for which the $C_r(\lambda')$ curve would have a local maximum in all layers. However, if we focus on the first eight layers in the 153–293 km depth range in which the correlations were sufficiently large in the previous tests, the $C_r(\lambda')$ curve has local maxima in five layers at $\lambda' = 5 \times 10^{-3}$ which verifies our choice of the optimal regularization parameter (Fig. 5).

The correlation profiles of conductivity structures computed using the Md60p1j1, Md60p3j1, Md60p5j1, Md60p5j31, Md60p5j21 and Md60p5j11 configurations are depicted in Fig. 7. The confidence intervals are not plotted since they are reasonably narrow as shown in the previous tests. We discuss the individual profiles in the following paragraphs starting with the lowest-correlation profile Md60p5j1 and finishing with the highest-correlation profile Md60p5j31.

In the Md60p5j1 configuration, the maximum Global-Ocean correlation coefficient is as low as 0.27. The Pacific Ocean correlation coefficients are even smaller in absolute value and negative in most layers. The best correlation is achieved beneath the Atlantic Ocean where correlation coefficients are larger than 0.4 in some layers but all of them are smaller than 0.5. The 5 per cent noise added to all spherical harmonic coefficients of input data seriously corrupted the reconstructed conductivity.

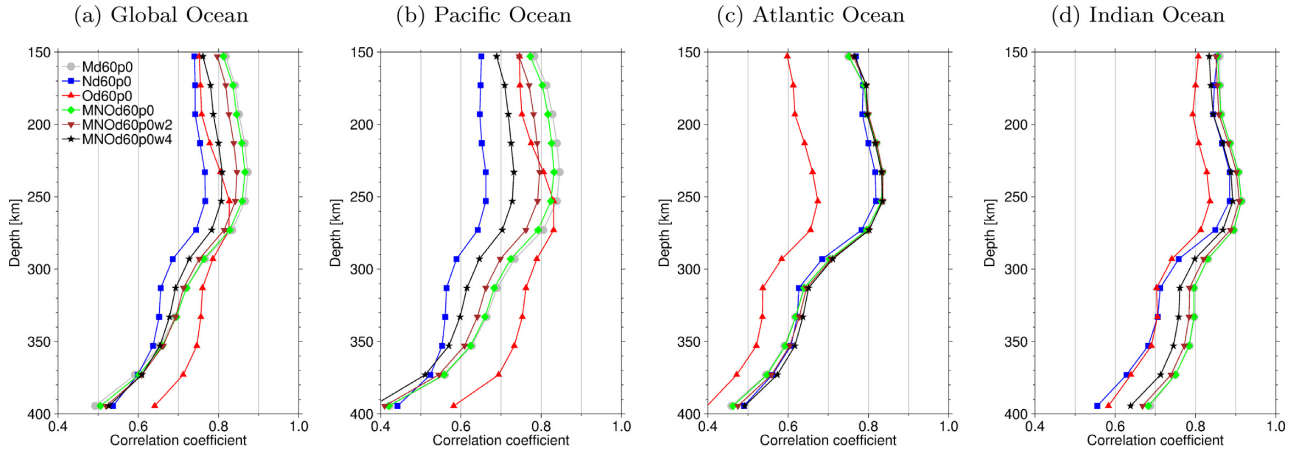


Figure 8. Impact of the different tidal constituents on the depth profiles of correlation coefficient beneath the (a) Global Ocean, (b) Pacific Ocean, (c) Atlantic Ocean and (d) Indian Ocean computed in the Nd60p0, Od60p0, MNod60p0, MNod60p0w2 and MNod60p0w4 tests.

In the Md60p3j1 configuration, the Global-Ocean correlation coefficients are still poor. The correlation beneath the Atlantic Ocean is more favourable. The Atlantic-Ocean correlation coefficients are larger than 0.5 in the 213–273 km depth range with the maximum of 0.59. Such accuracy might be acceptable. Indeed, the spatial distribution of conductivity anomalies is approximately correct (not shown).

The Md60p5j11 and Md60p1j1 Global-Ocean correlation profiles are similar. The correlation coefficients are larger than 0.7 in the 233–253 and 193–273 km depth ranges in the Md60p5j11 and Md60p1j1 Global-Ocean correlation profiles, respectively. The average Md60p5j11 and Md60p1j1 Global-Ocean correlation coefficients are 0.56 and 0.60 which corresponds to the decrease of 0.20 and 0.16 with respect to the reference noise-free test, respectively. In terms of regional variability, the correlations between the target model and the Md60p5j11 or Md60p1j1 conductivity structures are comparable beneath the Pacific Ocean. However, the reconstructed Md60p1j1 conductivity is closer to the target conductivity than the Md60p5j11 conductivity beneath the Atlantic and Indian Oceans. In fact, the Md60p1j1 correlation profile beneath the Atlantic Ocean is close to the Md60p5j21 profile.

As expected, the recovered Md60p5j21 and Md60p5j31 conductivity models are the closest to the target model. The Md60p5j21 and Md60p5j31 correlation profiles have similar shapes as the reference noise-free correlation profiles beneath the Global Ocean as well as beneath the individual ocean basins. The average Md60p5j21 and Md60p5j31 Global-Ocean correlation coefficients are 0.72 and 0.75 which is comparable to the noise-free $j_{\max D} = 30$ and $j_{\max D} = 60$ Global-Ocean correlation coefficients, respectively.

4.4 Noise-free N_2 and O_1 synthetic data with truncation degree $j_{\max D} = 60$

In addition to the M_2 TIMF, Sabaka *et al.* (2020) and Grayver & Olsen (2019) recovered also the TIMFs of the larger lunar elliptic semidiurnal tide N_2 and the lunar diurnal tide O_1 . Grayver & Olsen (2019) extracted the TIMF of N_2 and O_1 tidal constituents up to the truncation degree 12. Sabaka *et al.* (2020) used the truncation degree 36 but the extracted TIMFs are provided only up to degree 18. According to Sabaka *et al.* (2020), there is a good agreement between the large-scale features of TIMFs of Sabaka *et al.* (2020) and Grayver & Olsen (2019) but the correlation between the respective TIMFs decreases from M_2 to N_2 and O_1 .

In order to evaluate the performance of N_2 and O_1 constituents in the inverse problem, we execute the Nd60p0, Od60p0 and MNod60p0 noise-free tests with $j_{\max D} = 60$ in which we use N_2 , O_1 and combined $M_2 + N_2 + O_1$ TIMFs, respectively. The corresponding correlation profiles are depicted in Fig. 8. All three solutions are globally acceptable. However, the MNod60p0 solution which combines three tidal constituents is more distant from the target model than the Md60p0 solution with some exceptions in the deep layers.

The M_2 signal reaches up to 2 nT at 430 km altitude whereas N_2 and O_1 signals are weaker, they hardly reach 0.5 nT (Grayver & Olsen 2019). We perform two additional tests, MNod60p0w2 and MNod60p0w4 in which we use $w^{M_2} = 1$ (nT)⁻², $w^{N_2} = w^{O_1} = 2$ (nT)⁻² and $w^{M_2} = 1$ (nT)⁻², $w^{N_2} = w^{O_1} = 4$ (nT)⁻², respectively, to compensate the lower amplitudes of N_2 and O_1 signals. The respective correlation profiles in Fig. 8 show that the correlation decreases with increasing w^{N_2} and w^{O_1} weights.

We argue that there are two reasons why N_2 and O_1 tidal constituents do not improve the inversion. First, the N_2 and O_1 solutions depart from the target model beneath the Pacific and Atlantic Oceans, respectively. Secondly, the N_2 TIMF do not provide new pieces of information since the N_2 frequency 12.66 hr is close to the M_2 frequency 12.42 hr and the spatial patterns of N_2 and M_2 TIMFs are similar (figs 8 and 9 in Sabaka *et al.* 2020).

4.5 Sensitivity analysis

We complete our study with an analysis of sensitivity of misfit to the change of certain model parameters. For this purpose, we compute the Hessian for the target model and the $6^\circ \mathcal{M}_a$ mask. Since the number of model parameters is large in our discretization, we modify the

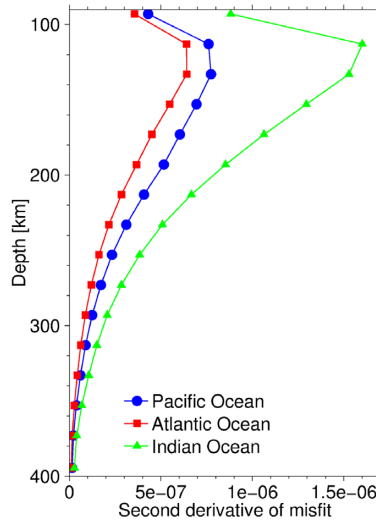


Figure 9. Sensitivity analysis of misfit based on the depth profiles of pure second derivative beneath the Pacific, Atlantic and Indian Oceans computed in the target model.

procedure described in Appendix B in order to decrease the size of Hessian. We replace the derivative D_α (and D_β) with respect to the α th model parameter with the directional derivative $\mathbf{v} \cdot \mathbf{D}_\mathbf{m}$ (and $\mathbf{v}' \cdot \mathbf{D}_\mathbf{m}$), where $\mathbf{v}$ is a prescribed unit vector in the M -dimensional model space. In each layer where we reconstruct the conductivity structure, we consider three vectors in the directions of the Pacific, Atlantic and Indian Oceans. For example, the α th component of the Pacific vector is equal to $(\mathbf{v}_P)_\alpha = 1/\sqrt{M_P}$, where M_P is the number of blocks in the Pacific subset of the mask $\mathcal{M}_\alpha$, if the α th model parameter affects the conductivity beneath the Pacific Ocean in that layer otherwise $v_\alpha = 0$. The size of the modified Hessian is $(3N_l) \times (3N_l)$ which is a significant reduction of the size $(1143N_l) \times (1143N_l)$ of the original Hessian. The sensitivity analysis is then less computationally demanding but it is also less detailed since it provides the sensitivity of misfit to a set of parameters rather than to a single parameter.

The diagonal of Hessian computed in the target model in which we extended the 3-D upper mantle up to 83 km depth is shown in Fig. 9. It is plotted as a depth profile of the pure (not mixed) second derivatives of misfit. All pure second derivatives are positive since the target model is a point in the model space where misfit has its minimum (all first derivatives are zero). In all three global oceans, there is an evident decrease of the second derivative with depth. The non-diagonal terms in the Hessian behave similarly (not shown). As a consequence, the reconstructed conductivity model is also expected to be more distant from the target conductivity model with increasing depth. In particular, the second derivative drops below 20 per cent of its maximum value in the profile at 253 km depth beneath the Atlantic and Indian Oceans and at 293 km depth beneath the Pacific Ocean. Of course, the 20 per cent threshold is *ad hoc* chosen and the behaviour of misfit at one point of model space can be misleading.

5 CONCLUSIONS

We presented a new effective approach to the global 3-D frequency-domain EM inverse problem of tidally induced magnetic field (TIMF). We tested the approach in a series of synthetic 3-D inverse problems. We have four main conclusions:

- (i) The 3-D sub-oceanic upper-mantle conductivity structure can be reconstructed from satellite TIMF data and it is closer to the target conductivity in the approximate 150–270 km depth range than in the 270–400 km depth range which is supported by the sensitivity analysis based on the diagonal of Hessian matrix.
- (ii) Data quality might be the limiting factor for the 3-D inversion of satellite-observed TIMF. In particular, the level of noise on the lowest spherical harmonic degrees is critical. The reconstructed conductivity has been seriously corrupted by the 3 per cent Gaussian noise added to all data coefficients in the Md60p3j1 test.
- (iii) The truncation degree of modern TIMF data products is sufficiently high for the 3-D inversion as demonstrated in the Md24p0 and Md30p0 tests.
- (iv) The additional N_2 and O_1 tidal constituents do not improve the reconstructed conductivity as demonstrated in the MNOd60p0, MNOd60p0w2 and MNOd60p0w4 tests.

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DATA AVAILABILITY

The results of the forward and inverse modelling will be shared on reasonable request to the corresponding author.

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APPENDIX A: FORMULATION OF THE ADJOINT PROBLEM

The quasi-Newton algorithm used in the minimization of the penalty function in eq. (14) requires repeated calculations of the M -dimensional gradients $\mathbf{D}_m \chi^2(\mathbf{m})$,

$$\mathbf{D}_m \chi^2(\mathbf{m}) = (D_1 \chi^2(\mathbf{m}), D_2 \chi^2(\mathbf{m}), \dots, D_M \chi^2(\mathbf{m})), \quad (\text{A1})$$

where D_α is the derivative with respect to the α th model parameter, and $\mathbf{D}_m R^2(\mathbf{m})$ (defined analogously) in subsequent points $\mathbf{m}_k \in \mathbb{R}^M$, as the iterative scheme proceeds. While the derivative of the regularization term can be expressed analytically, the adjoint problem has to be defined and solved for an efficient calculation of the data misfit derivative. Maksimov & Velimský (2017) have carried out the detailed derivation for the spherical-harmonic discretization of the EMI equation in the time domain. Here we extend their results into the frequency domain, avoiding one integration by parts over the time variable, but having to deal in addition with the complex fields and complex spherical harmonic base.

We start with the quasi-stationary Maxwell equations from which the EMI eq. (1) was derived, and introduce the equivalent relations for the adjoint magnetic and electric fields $\hat{\mathbf{B}}$, $\hat{\mathbf{E}}$ of the tidal mode l ,

$$\rho \operatorname{curl} \mathbf{B} = \mu_0 (\mathbf{E} + \mathbf{u} \times \mathbf{B}_M), \quad (\text{A2})$$

$$\operatorname{curl} \mathbf{E} = i\omega_l \mathbf{B}, \quad (\text{A3})$$

$$\rho \operatorname{curl} \hat{\mathbf{B}} = \mu_0 \hat{\mathbf{E}}, \quad (\text{A4})$$

$$\operatorname{curl} \hat{\mathbf{E}} = -i\omega_l \hat{\mathbf{B}}, \quad (\text{A5})$$

where $\rho(\mathbf{r})$ denotes the electrical resistivity, $\rho = 1/\sigma$, introduced here for the sake of formula compactness. Note the absence of the motional forcing in the adjoint Ampère's law (A4), and change of the sign for the angular frequency in the adjoint version of the Faraday's law (A5). The adjoint EMI equation is then readily derived, and complex conjugation is applied to it,

$$\operatorname{curl} [\rho(\mathbf{r}) \operatorname{curl} \hat{\mathbf{B}}(\mathbf{r}; \omega_l)] + i\omega_l \mu_0 \hat{\mathbf{B}}(\mathbf{r}; \omega_l) = \mathbf{0}, \quad (\text{A6})$$

$$\operatorname{curl} [\rho(\mathbf{r}) \operatorname{curl} \hat{\mathbf{B}}(\mathbf{r}; \omega_l)] - i\omega_l \mu_0 \hat{\mathbf{B}}(\mathbf{r}; \omega_l) = \mathbf{0}. \quad (\text{A7})$$

The complex-conjugate adjoint EMI eq. (A7) is formally identical to the forward EMI eq. (1) without the motional forcing term.

Let us now apply the derivative D_α to the EMI eq. (1) for a particular tidal mode l , multiply it by the complex-conjugated adjoint magnetic field $\hat{\mathbf{B}}$, and integrate over the sphere G . Since the motional terms on the right-hand side are independent of the model parameters $\mathbf{m}$, the result is zero,

$$\mathcal{I}_l = \int_G \hat{\mathbf{B}}(\omega_l) \cdot D_\alpha \{ \operatorname{curl} [\rho \operatorname{curl} \mathbf{B}(\omega_l)] - i\omega_l \mu_0 \mathbf{B}(\omega_l) \} dV = 0. \quad (\text{A8})$$

This expression is now modified with the use of eqs (A2–A5, A7), and a vector identity

$$\int_G \mathbf{g} \cdot \operatorname{curl} \mathbf{f} dV = \int_G \mathbf{f} \cdot \operatorname{curl} \mathbf{g} dV + \int_{\partial G} (\mathbf{f} \times \mathbf{g}) \cdot \mathbf{e}_r dS, \quad (\text{A9})$$

in order to eliminate the $D_\alpha \mathbf{B}$ term. Omitting explicit notation of the dependence on ω_l , we can write

$$\begin{aligned} \mathcal{I}_l &= \int_G \hat{\mathbf{B}} \cdot D_\alpha [\mu_0 \operatorname{curl} (\mathbf{E} + \mathbf{u} \times \mathbf{B}_M) - i\omega_l \mu_0 \mathbf{B}] dV \\ &= \int_G \hat{\mathbf{B}} \cdot [\mu_0 \operatorname{curl} D_\alpha \mathbf{E} - i\omega_l \mu_0 D_\alpha \mathbf{B}] dV \\ &= \int_G \mu_0 \operatorname{curl} \hat{\mathbf{B}} \cdot D_\alpha \mathbf{E} dV + \int_{\partial G} \mu_0 (D_\alpha \mathbf{E} \times \hat{\mathbf{B}}) \cdot \mathbf{e}_r dS - \int_G i\omega_l \mu_0 \hat{\mathbf{B}} \cdot D_\alpha \mathbf{B} dV \\ &= \int_G D_\alpha \rho \operatorname{curl} \hat{\mathbf{B}} \cdot \operatorname{curl} \mathbf{B} dV + \int_G \operatorname{curl} (\rho \operatorname{curl} \hat{\mathbf{B}}) \cdot D_\alpha \mathbf{B} dV - \int_G i\omega_l \mu_0 \hat{\mathbf{B}} \cdot D_\alpha \mathbf{B} dV \\ &\quad + \int_{\partial G} \mu_0 (D_\alpha \mathbf{B} \times \hat{\mathbf{E}}) \cdot \mathbf{e}_r dS + \int_{\partial G} \mu_0 (D_\alpha \mathbf{E} \times \hat{\mathbf{B}}) \cdot \mathbf{e}_r dS \\ &= \int_G D_\alpha \rho \operatorname{curl} \hat{\mathbf{B}} \cdot \operatorname{curl} \mathbf{B} dV + \int_G [\operatorname{curl} (\rho \operatorname{curl} \hat{\mathbf{B}}) - i\omega_l \mu_0 \hat{\mathbf{B}}] \cdot D_\alpha \mathbf{B} dV \\ &\quad + \int_{\partial G} \mu_0 (D_\alpha \mathbf{B} \times \hat{\mathbf{E}} + D_\alpha \mathbf{E} \times \hat{\mathbf{B}}) \cdot \mathbf{e}_r dS \\ &= \int_G D_\alpha \rho \operatorname{curl} \hat{\mathbf{B}}(\omega_l) \cdot \operatorname{curl} \mathbf{B}(\omega_l) dV + \mathcal{I}_{S,l}. \end{aligned} \quad (\text{A10})$$

The term $D_\alpha \mathbf{B}$ has been successfully cancelled from the volume integral, but it remains to rearrange the surface integral $\mathcal{I}_{S,l}$. For this purpose, we need to introduce the spherical harmonic vectors for $j = 1, \dots, \infty$, $m = -j, \dots, +j$,

$$\mathcal{S}_{jm}^{(-1)}(\Omega) = Y_{jm}(\Omega) \mathbf{e}_r, \quad (\text{A11})$$

$$\mathcal{S}_{jm}^{(+1)}(\Omega) = \operatorname{grad}_\Omega Y_{jm}(\Omega), \quad (\text{A12})$$

$$\mathcal{S}_{jm}^{(0)}(\Omega) = \mathbf{e}_r \times \operatorname{grad}_\Omega Y_{jm}(\Omega), \quad (\text{A13})$$

corresponding, respectively, to the poloidal vertical, poloidal horizontal, and toroidal components of a vector, and forming a complex orthogonal base on the unit sphere with norm

$$\int_{\Omega} \bar{\mathcal{S}}_{jm}^{(\lambda)} \mathcal{S}_{j'm'}^{(\lambda')} d\Omega = \delta_{jj'} \delta_{mm'} \delta_{\lambda\lambda'} [\delta_{-1\lambda} + \Pi_j (1 - \delta_{-1\lambda})], \quad (\text{A14})$$

where $\Pi_j = j(j+1)$. The individual vector fields are expanded into series of spherical harmonic coefficients,

$$\mathbf{B}(\mathbf{r}, \omega_l) = \sum_{j=1}^{j_{\max}} \sum_{m=-j}^j \sum_{\lambda=-1}^{+1} E_{jm}^{(\lambda)}(r; \omega_l) \mathcal{S}_{jm}^{(\lambda)}(\Omega), \quad (\text{A15})$$

$$\mathbf{E}(\mathbf{r}, \omega_l) = \sum_{j=1}^{j_{\max}} \sum_{m=-j}^j \sum_{\lambda=-1}^{+1} E_{jm}^{(\lambda)}(r; \omega_l) \mathcal{S}_{jm}^{(\lambda)}(\Omega), \quad (\text{A16})$$

$$\hat{\mathbf{B}}(\mathbf{r}, \omega_l) = \sum_{j=1}^{j_{\max}} \sum_{m=-j}^j \sum_{\lambda=-1}^{+1} \hat{E}_{jm}^{(\lambda)}(r; \omega_l) \mathcal{S}_{jm}^{(\lambda)}(\Omega), \quad (\text{A17})$$

$$\hat{\mathbf{E}}(\mathbf{r}, \omega_l) = \sum_{j=1}^{j_{\max}} \sum_{m=-j}^j \sum_{\lambda=-1}^{+1} \hat{E}_{jm}^{(\lambda)}(r; \omega_l) \mathcal{S}_{jm}^{(\lambda)}(\Omega). \quad (\text{A18})$$

In particular, expressing the radial components of the Faraday's law yields, respectively, for the spherical harmonic coefficients of electric and magnetic field, and their adjoint counterparts,

$$\frac{\Pi_j}{r} E_{jm}^{(0)} = -i\omega B_{jm}^{(-1)}, \quad (\text{A19})$$

$$\frac{\Pi_j}{r} \hat{E}_{jm}^{(0)} = +i\omega \hat{B}_{jm}^{(-1)}. \quad (\text{A20})$$

The boundary condition (3) in the spherical harmonic base reads as

$$B_{jm}^{(0)}(a) = 0, \quad (\text{A21})$$

$$B_{jm}^{(-1)}(a) = -j G_{jm}^{(e)} + (j+1) G_{jm}^{(i)} = (j+1) G_{jm}^{(i)}, \quad (\text{A22})$$

$$B_{jm}^{(+1)}(a) = -G_{jm}^{(e)} - G_{jm}^{(i)} = -G_{jm}^{(i)}, \quad (\text{A23})$$

and similarly, for the adjoint fields,

$$\hat{B}_{jm}^{(0)}(a) = 0, \quad (\text{A24})$$

$$\hat{B}_{jm}^{(-1)}(a) = -j \hat{G}_{jm}^{(e)} + (j+1) \hat{G}_{jm}^{(i)}, \quad (\text{A25})$$

$$\hat{B}_{jm}^{(+1)}(a) = -\hat{G}_{jm}^{(e)} - \hat{G}_{jm}^{(i)}, \quad (\text{A26})$$

where $\hat{G}_{jm}^{(e)}$ and $\hat{G}_{jm}^{(i)}$ are, respectively, the external and internal spherical harmonic coefficients of a scalar adjoint magnetic potential $\hat{U}$ in the insulating atmosphere.

Equipped with the spherical harmonic formalism, we proceed with evaluation of the surface integral from eq. (A10),

$$\begin{aligned} \mathcal{I}_{S,l} &= \mu_0 a^2 \sum_{jm} \Pi_j \left[D_\alpha B_{jm}^{(1)}(a) \hat{E}_{jm}^{(0)}(a) - D_\alpha B_{jm}^{(0)}(a) \hat{E}_{jm}^{(1)}(a) + \hat{B}_{jm}^{(0)}(a) D_\alpha E_{jm}^{(1)}(a) - \hat{B}_{jm}^{(1)}(a) D_\alpha E_{jm}^{(0)}(a) \right] \\ &= i\omega_l \mu_0 a^3 \sum_{jm} \left[-D_\alpha B_{jm}^{(1)}(a) \hat{B}_{jm}^{(-1)}(a) + \hat{B}_{jm}^{(1)}(a) D_\alpha B_{jm}^{(-1)}(a) \right] \\ &= i\omega_l \mu_0 a^3 \sum_{jm} \left\{ D_\alpha G_{jm}^{(i)} \left[-j \hat{G}_{jm}^{(e)} + (j+1) \hat{G}_{jm}^{(i)} \right] - \left(\hat{G}_{jm}^{(e)} + \hat{G}_{jm}^{(i)} \right) (j+1) D_\alpha G_{jm}^{(i)} \right\} \\ &= -i\omega_l \mu_0 a^3 \sum_{jm} (2j+1) D_\alpha G_{jm}^{(i)}(\omega_l) \hat{G}_{jm}^{(e)}(\omega_l). \end{aligned} \quad (\text{A27})$$

We have reduced the surface integral to a single spherical harmonic series, however, it still depends of $D_\alpha G_{jm}^{(i)}$.

Finally, we apply the D_α operator directly to the misfit definition (8), and proceed with cancellation of $D_\alpha G_{jm}^{(i)}$ term. This is obtained by addition of the sum of integrals $\mathcal{I}_l = 0$ and their complex conjugates over all tidal modes (scaled by a constant to keep the expressions

dimensionless), and rearranging terms,

$$\begin{aligned}
D_\alpha \chi^2(\mathbf{m}) &= \sum_l \sum_{jm} \sum_{j'm'} N_{jj'} \left(\frac{a}{r_S} \right)^{\frac{i+j'}{2}} \left\{ \left[\bar{G}_{jm}^{(i)}(\omega_l) - \bar{G}_{jm}^{(i, \text{obs})}(\omega_l) \right] W_{jmj'm'}^l D_\alpha G_{j'm'}^{(i)}(\omega_l) \right. \\
&\quad \left. + D_\alpha \bar{G}_{jm}^{(i)}(\omega_l) W_{jmj'm'}^l \left[G_{j'm'}^{(i)}(\omega_l) - G_{j'm'}^{(i, \text{obs})}(\omega_l) \right] \right\} + \sum_l \frac{1}{4\pi a^3 \omega_l \mu_0} (\mathcal{I}_l + \bar{\mathcal{I}}_l) \\
&= \frac{2}{4\pi a^3 \mu_0} \sum_l \frac{1}{\omega_l} \text{Re} \left\{ \int_G D_\alpha \rho \text{curl} \hat{\mathbf{B}}(\omega_l) \cdot \text{curl} \mathbf{B}(\omega_l) dV \right\} \\
&\quad + \sum_l \sum_{j'm'} \left\{ \sum_{jm} N_{jj'} \left(\frac{a}{r_S} \right)^{\frac{i+j'}{2}} \left[\bar{G}_{jm}^{(i)}(\omega_l) - \bar{G}_{jm}^{(i, \text{obs})}(\omega_l) \right] W_{jmj'm'}^l \right. \\
&\quad \left. - \frac{i}{4\pi} (2j' + 1) \hat{G}_{j'm'}^{(e)}(\omega_l) \right\} D_\alpha G_{j'm'}^{(i)}(\omega_l) \\
&\quad + \sum_l \sum_{jm} \left\{ \sum_{j'm'} N_{jj'} \left(\frac{a}{r_S} \right)^{\frac{i+j'}{2}} \left[G_{j'm'}^{(i)}(\omega_l) - G_{j'm'}^{(i, \text{obs})}(\omega_l) \right] \bar{W}_{j'm'jm}^l \right. \\
&\quad \left. + \frac{i}{4\pi} (2j' + 1) \hat{G}_{jm}^{(e)}(\omega_l) \right\} D_\alpha \bar{G}_{jm}^{(i)}(\omega_l) \\
&= \frac{2}{4\pi a^3 \mu_0} \sum_l \frac{1}{\omega_l} \text{Re} \left\{ \int_G D_\alpha \rho(\mathbf{m}) \text{curl} \hat{\mathbf{B}}(\mathbf{m}; \omega_l) \cdot \text{curl} \mathbf{B}(\mathbf{m}; \omega_l) dV \right\}. \tag{A28}
\end{aligned}$$

The cancellation of the terms depending on $D_\alpha G_{j'm'}^{(i)}$ and $D_\alpha \bar{G}_{jm}^{(i)}$ is made possible by postulating the non-zero external boundary condition on the adjoint magnetic field,

$$\hat{G}_{j'm'}^{(e)}(\omega_l) = -\frac{4\pi i}{2j' + 1} \sum_{jm} N_{jj'} \left(\frac{a}{r_S} \right)^{\frac{i+j'}{2}} \left[\bar{G}_{jm}^{(i)}(\mathbf{m}; \omega_l) - \bar{G}_{jm}^{(i, \text{obs})}(\omega_l) \right] W_{jmj'm'}^l, \tag{A29}$$

complementing the adjoint EMI eq. (A7). The adjoint problem is excited by virtual sources, external to the insulating atmosphere, whose strength is proportional to the weighted residua of the forward problem calculated at point $\mathbf{m}$ in the model space. The gradient of the data misfit in the model space is then obtained by combining the curl of the forward and adjoint solutions with the gradient of the conductivity model. Calculation of $D_\alpha \rho(\mathbf{m})$ from eq. (7) is trivial, bearing in mind that $\log_{10}(\rho/\rho_0) = -\log_{10}(\sigma/\sigma_0)$. It also implies that for each component m_α , the integration in eq. (A28) will span only the corresponding block $I_\alpha \subset G$.

APPENDIX B: HESSIAN OF THE MISFIT FUNCTION

Following the time-domain derivation by Maksimov & Velínský (2017), we introduce here the application of the adjoint solution to fast calculation of the misfit Hessian for the frequency-domain, tidally driven EMI problem. The matrix of second derivatives of the data misfit is constructed by differentiating directly the misfit gradient (A28),

$$\begin{aligned}
D_{\alpha\beta} \chi^2(\mathbf{m}) &= \frac{2}{4\pi a^3 \mu_0} \sum_l \frac{1}{\omega_l} \text{Re} \left\{ \int_G D_{\alpha\beta} \rho(\mathbf{m}) \text{curl} \hat{\mathbf{B}}(\mathbf{m}; \omega_l) \cdot \text{curl} \mathbf{B}(\mathbf{m}; \omega_l) dV \right\} \\
&\quad + \frac{2}{4\pi a^3 \mu_0} \sum_l \frac{1}{\omega_l} \text{Re} \left\{ \int_G D_\alpha \rho(\mathbf{m}) \text{curl} \hat{\mathbf{B}}(\mathbf{m}; \omega_l) \cdot \text{curl} D_\beta \mathbf{B}(\mathbf{m}; \omega_l) dV \right\} \\
&\quad + \frac{2}{4\pi a^3 \mu_0} \sum_l \frac{1}{\omega_l} \text{Re} \left\{ \int_G D_\alpha \rho(\mathbf{m}) \text{curl} D_\beta \hat{\mathbf{B}}(\mathbf{m}; \omega_l) \cdot \text{curl} \mathbf{B}(\mathbf{m}; \omega_l) dV \right\}. \tag{B1}
\end{aligned}$$

The first integral requires a simple evaluation of $D_{\alpha\beta} \rho(\mathbf{m})$ obtained directly from eq. (7), and the forward and adjoint fields $\mathbf{B}(\mathbf{m}; \omega_l)$, $\hat{\mathbf{B}}(\mathbf{m}; \omega_l)$. The second and third integrals require the calculation of the forward and adjoint scatter fields $D_\alpha \mathbf{B}(\mathbf{m}; \omega_l)$, $D_\alpha \hat{\mathbf{B}}(\mathbf{m}; \omega_l)$, respectively.

The forward scatter equation is obtained by differentiating the forward EMI problem (1),

$$\text{curl} [\rho(\mathbf{r}) \text{curl} D_\alpha \mathbf{B}(\mathbf{r}; \omega_l)] - i\omega_l \mu_0 D_\alpha \mathbf{B}(\mathbf{r}; \omega_l) = -\text{curl} [D_\alpha \rho(\mathbf{r}) \text{curl} \mathbf{B}(\mathbf{r}; \omega_l)], \tag{B2}$$

with a boundary condition

$$D_\alpha G_{jm}^{(e)} = 0. \quad (\text{B3})$$

The adjoint scatter equation is then obtained by differentiating the adjoint problem (A7),

$$\text{curl} \left[\rho(\mathbf{r}) \text{curl} D_\alpha \hat{\mathbf{B}}(\mathbf{r}; \omega_l) \right] - i\omega\mu_0 D_\alpha \hat{\mathbf{B}}(\mathbf{r}; \omega_l) = -\text{curl} \left[D_\alpha \rho(\mathbf{r}) \text{curl} \hat{\mathbf{B}}(\mathbf{r}; \omega_l) \right], \quad (\text{B4})$$

with a boundary condition obtained from eq. (A29),

$$D_\alpha \hat{G}_{j'm'}^{(e)}(\omega_l) = -\frac{4\pi i}{2j'+1} \sum_{jm} N_{jj'} \left(\frac{a}{r_s} \right)^{\frac{j+j'}{2}} D_\alpha \overline{G}_{jm}^{(i)}(\mathbf{m}; \omega_l) W_{jmj'm'}^l, \quad (\text{B5})$$

where $D_\alpha \overline{G}_{jm}^{(i)}(\mathbf{m}; \omega_l)$ are the induced SH coefficients of the forward scatter problem. Note, that the calculation of the full Hessian requires $2M + 2$ solutions of the EMI equation: one forward, one adjoint, M forward scatter and M adjoint scatter problems. All four types of problem use the same ElmgFD solver, and should be calculated in this order because of the dependencies in the source and boundary terms.